

Crab's Physiological Stress Response to Acute Anthropogenic Noise Pollution in the Near Shore Environment

Abstract

Anthropogenic noise pollution in marine environments has increased substantially over recent decades and is an understudied stressor for nearshore invertebrate communities. Crustaceans detect vibration and low-frequency sound through specialized mechanoreceptors, and prior work has shown noise exposure can disrupt behavior and elevate stress responses in crabs. Puget Sound crab populations face this understudied stressor while already under significant pressure from the invasive European Green Crab (*Carcinus maenas*). Here we show that acute anthropogenic noise exposure induces a suppressed or torpor-like physiological state in Hairy Shore Crabs (*Hemigrapsus oregonensis*). Using resazurin fluorescence, terrestrial righting time, and hemolymph glucose we found that treatment crabs exposed to ~75 dB of electric dance music for two hours per week, for three consecutive weeks showed lower oxygen consumption, longer righting times, and greater variation in hemolymph glucose than control crabs. Despite no assay resulting in a statistical significance with an $\alpha = 0.05$ the visual trends were interesting and warrant further investigation. This torpor-like response could be damaging during sensitive life history periods such as spawning and they can potentially compound existing population pressures on Dungeness crab (*Metacarcinus magister*). These results warrant further investigation with larger sample sizes and direct testing on dungeness crab to inform conservation strategy and noise pollution legislation in the Puget Sound nearshore environment.

Intro

Washington State's ecologically, culturally, and economically important Dungeness crab (*Metacarcinus magister*) fishery has experienced an extreme decline. Harvests from state and tribal fisheries in the South Puget Sound peaked at 289,500 pounds in 2012 before plummeting to 9,500 pounds in 2017 (Dufault & Bosley, 2024, as cited in Good, 2025). These populations are facing increasing pressures from the invasive European green crab (*Carcinus maenas*), ocean acidification, and other environmental stressors (Jensen et al. 2002; Fisher et al. 2024; Miller et al. 2016). However, less attention has been paid to the physiological impacts of anthropogenic noise pollution on these invertebrates (Aimon et al. 2021). Studies have begun to address this knowledge gap and have identified an increase in aquatic invertebrate behavioral and physiological stress responses in juvenile invertebrates when exposed to artificial noise (Pani et al. 2026). Additionally, research has shown that anthropogenic noise has a significant negative effect on invertebrate physiology (Davies et al. 2024).

Anthropogenic noise in marine environments has grown substantially, with intensity levels approximately doubling about every decade. This pollution is driven by global shipping, recreational boating, any on or near water activity and expanding coastal development (Jalkanen et al. 2022). The Port of Seattle has recognized this and implemented underwater acoustic

monitoring systems to track local noise (Holt et al. 2026). Crustaceans detect these vibrations through specialized mechanoreceptors, and previous studies have indicated that noise exposure can elevate stress hormones, impair locomotion, and disrupt vital behaviors in various invertebrates (Aimon et al. 2021). The specific effects of noise on Pacific Northwest crab populations remains unstudied.

To address this knowledge gap, we used the Hairy Shore Crab (*Hemigrapsus oregonensis*) as a model organism to measure physiological responses to acute noise pollution. We designed a pilot study simulating a nearshore or floating concert series. Crabs were exposed to approximately 75 dB of electronic dance music for two hours per week over three consecutive weeks. We asked whether acute noise exposure would produce measurable physiological stress responses. We specifically tracked respiration rate, behavioral impairment, and hemolymph glucose as indicators of stress.

Methods

All of the hairy shore crabs were collected from the Puget Sound. To ensure consistency, a total of 12 crabs were selected for the experiment and divided into two groups (n=6 per treatment). Selection was non-random to maintain a standardized sex ratio. Individuals were selected based on similar body masses, and gravid females were excluded to avoid metabolic variation associated with reproduction. Each crab was marked with an identifier on the carapace to track individuals throughout the experiment (yellow = control, blue = treatment). Crabs were housed in separate, aerated plastic tubs held at 35 ppt salinity and 10°C. These conditions best reflect ambient Puget Sound conditions. This minimizes non-treatment stress on the crabs. Standard husbandry practices included weekly feedings and the implementation of protective measures. For example, temporary isolation while individuals were molting.

Treatment crabs were exposed to a ~75 dB electric dance music playlist for two hours on a weekly basis for three weeks. During the first hour of exposure, crabs remained in their home tanks. For the second hour, crabs were transferred to well plates containing a resazurin. This allowed the measurement of respiration during the acute anthropogenic noise exposure. Resazurin samples were collected and processed according to the standardized protocol established by the Roberts Lab (Huffmyer et al. 2025)

Following the two-hour music exposure, each crab was subjected to a terrestrial righting-time test to measure impairment caused by noise. Crabs after treatment were one by one placed on their back, and timed as they flipped themselves over. Crabs were then rinsed in a two-stage saltwater bath to remove residual solvent before being returned to their respective holding tanks. In the final week of the experiment, after the standard testing protocol, we collected hemolymph samples from all individuals to quantify glucose levels using the Cayman Chemical Glucose Colorimetric Assay following the kits given procedure.

Results

We found that the hairy shore crab likely was physiologically impacted by the acute anthropogenic noise exposure but there were not significant statistical differences between control and treatment crabs, likely due to the small sample size.

During our experiment we experienced two mortalities and one successful molt, both happened in the control tank. The molted crab was quarantined and the animal was removed from subsequent testing. Combined, the mortality and molt, the control experiments in week two and three were performed with a smaller sample size.

Resazurin assay results, which measured the crab's respiratory rate, were statistically insignificant with an alpha of 0.05 when comparing the treatment to the control at all three timepoints (T0: p-value = 0.91, T1: p-value = 0.57, T2: p = 0.54). However, a consistent trend was observed across all three week's results (Figure 1, 2). Control crabs consumed more oxygen than treatment crabs at each timepoint. At T0 the treatment crabs on average had a fluorescence of 131.45 compared to the control's 111.47. This difference was negligible due to overlapping standard deviations, but it grew substantially by T1 (treatment: 1,345.63, control: 1,932.83) and T2 (treatment: 2,837.37, control: 4,357.68). A paired t-test across all timepoints and weeks approached but did not reach significance ($t = -2.20$, $df = 8$, $p\text{-value} = 0.059$). The small sample size and short term nature of the study resulted in non-significant statistical results, but the interesting visual trend warrants further future investigation.

Before outliers were evaluated and eventually removed, righting times appeared significantly longer in the treatment group. A Tukey outlier test confirmed two treatment data points as outliers. This was crab Blue 13 in Week 1 (46.81s) and Blue 12 in Week 3 (56.24s). Both crabs were exposed to the noise treatment. Although these data were removed from following statistical analysis, they biologically follow the expected trend and are important to our understanding of the treatment's effect on the crab's physiology. Following the outliers removal, no significant difference in righting time was detected between treatment (mean = 1.23s, $sd = 0.92$) and control crabs (mean = 1.06s, $sd = 1.29$) by a t-test ($t = -0.40$, $df = 19.2$, $p = 0.69$). However, there is an interesting visual trend in boxplots presented in figure 3. We see that the average righting time for the crabs exposed to the treatment is higher than the control crabs.

Finally, our glucose assay (which was not designed for crabs) resulted in a plot with no statistical significance but interesting visual trends. We noticed a larger range and distribution in the absorbance of the experimental crabs than we do when compared to the control group. With such a small sample size and an assay not designed for use on invertebrates a clear conclusion can not be drawn, but visually there is a trend of increased variation in hemolymph glucose in the treatment crabs than the control crabs.

Discussion

Although there were no statistically significant results, our findings still support the hypothesis that crabs exposed to nearshore acute anthropogenic noise pollution will have physiological changes and our results warrant further investigation with larger sample sizes and both hairy shore crabs and Dungeness crab.

We found that the treatment resulted in the crabs acting as if they were experiencing a torpor or forced suppressed state. We saw this with the consistent decrease in respiration in treatment crabs compared to controls across all three weeks, and treatment crabs showed increases in righting time. The righting time outliers suggests that while most crabs showed subtle suppression or physiological change, some individuals were much more severely affected. This artificially caused suppression could be damaging to wild crab populations if it occurred during an already high stress or sensitive period, for example during their spawning window. Exposure to acute anthropogenic noise could lead to a decrease in activity, thus a decrease in spawning success. This would be particularly damaging to an at-risk population like the Dungeness crab.

However, the increased variation in treatment crab hemolymph glucose opposes our hypothesis of a torpor-like state and matches a fight-or-flight response. Our new understanding is that the two-hour noise exposure exhausted the acute stress response, leaving crabs fatigued rather than activated. This pattern is consistent with Hapsari's meta-analysis of crustacean stress responses (Hapsari et al. 2025).

It is worth noting that our glucose assay was not designed or validated for crab hemolymph. The assay's results showed an extreme increase in hemolymph glucose variation when comparing the treatment crab's results to those of the controls. Biologically we expect the treatment crabs to have an elevated glucose level, as this is a known acute stress response in most animals. Future studies should use a crustacean specific assay to better capture metabolic stress induced glucose response.

With this pilot study's limited sample size and other confounding variables we recommend further investigation with larger sample sizes and the inclusion of Dungeness crab directly, to see if the trends become statistically significant. A future study will inform legislation around noise ordinances and conservation strategy in our nearshore environment and protect Puget Sound crab populations so that we can rebuild future populations into a viable fishery.

Figures

Figure 1. Resazurin assay fluorescence zeroed by blanks and crab weight over 30 minute time intervals for each week of experimentation. Orange lines represent control crabs and blue lines represent treatment crabs. Error bars represent $\pm$ one standard deviation. Y-axis scales change between weeks due to variation in fluorescence.

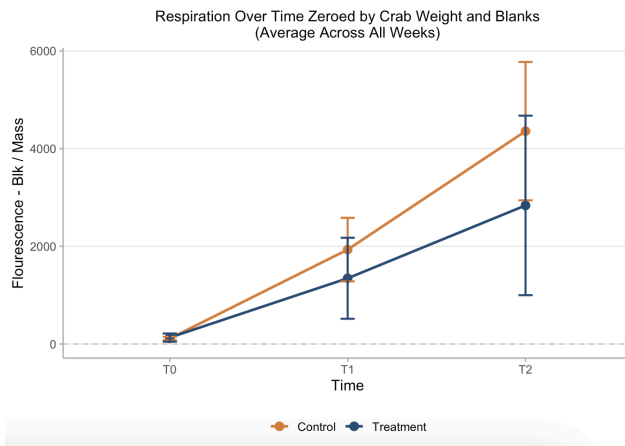
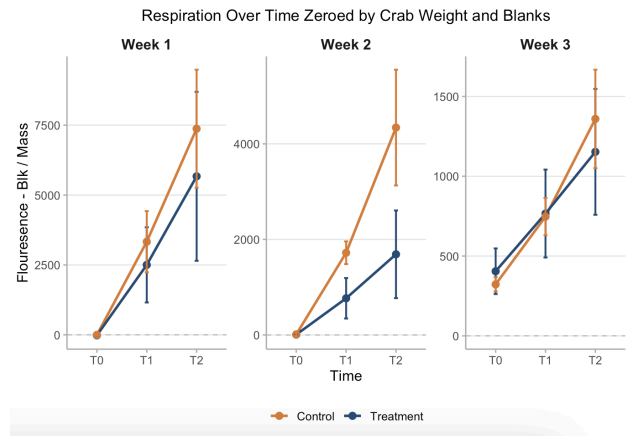


Figure 2. Mean resazurin assay fluorescence averaged across all three weeks of experimentation measured at three 30-minute intervals. Fluorescence values zeroed by blank and normalized by crab mass. The orange line represents control crabs and the blue line represents treatment crabs. Error bars are $\pm$ an average standard deviation across all three weeks.

Figure 3. Righting time in seconds for control and treatment crabs pooled across all three weeks of experimentation. Box represents the IQR, horizontal line indicates the median, and whiskers extend to 1.5x IQR. Individual data points are shown as filled circles. Two Tukey outliers from the treatment group (Blue 13 Week 1: 46.81s, Blue 12 Week 3: 56.24s) were removed.

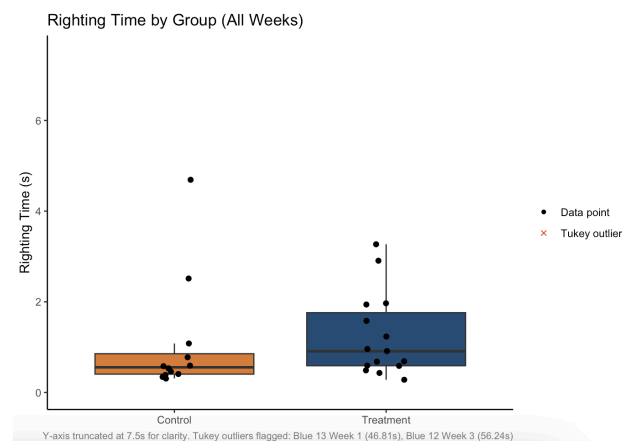
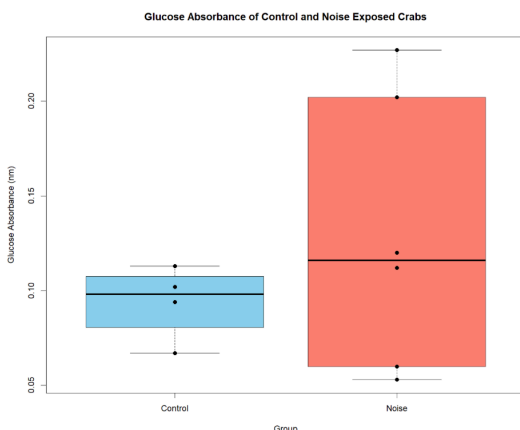


Figure 4. Glucose absorbance values from hemolymph samples collected in the final week of experimentation, separated by control and noise treatment groups. Absorbance was measured using the Cayman Chemical Glucose Colorimetric Assay. Box represents the IQR, horizontal line indicates the median, and whiskers extend to 1.5x IQR



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